NELLY: A transparent deep learning framework for patient-centric drug response prediction and prioritization

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1. Introduction

Over the past decade, several machine learning methods have been developed to predict patient responses to anti-cancer drugs. Despite methodological advances, they present limited translational relevance for personalized medicine.

The main limitations of current state-of-the-art methods can be summarized by:

1. **Inflated performance**
2. **Limited transparency**
3. **Scarce validation in patient-derived material**

2. Methods

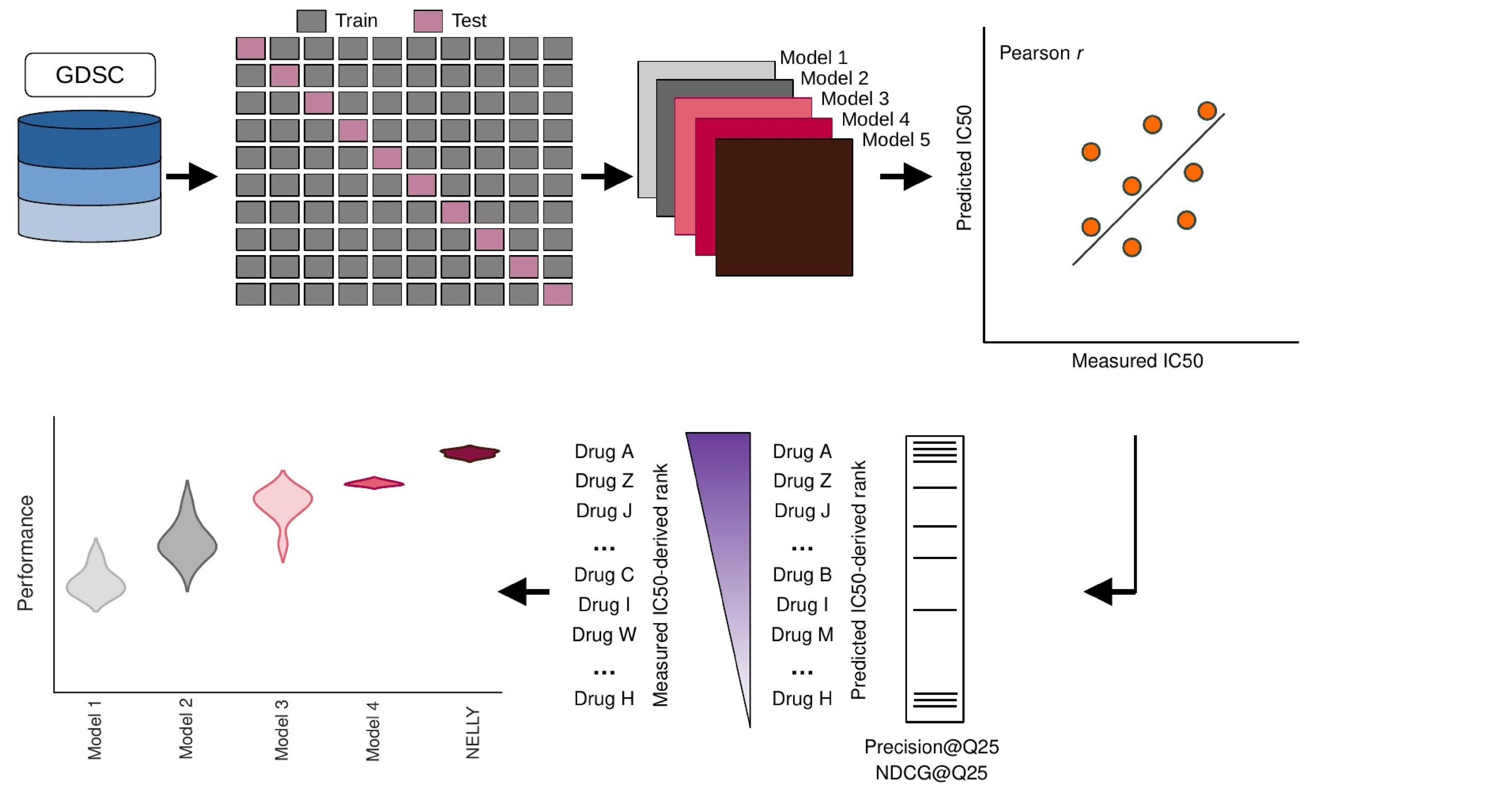

Figure 1. Translationally relevant benchmark for drug response prediction.
Study design applied for patient-centric model training and evaluation. Model performance was benchmarked based on Pearson correlation coefficient and ranking-based performance, namely Precision@Q25 and NDCG@Q25.

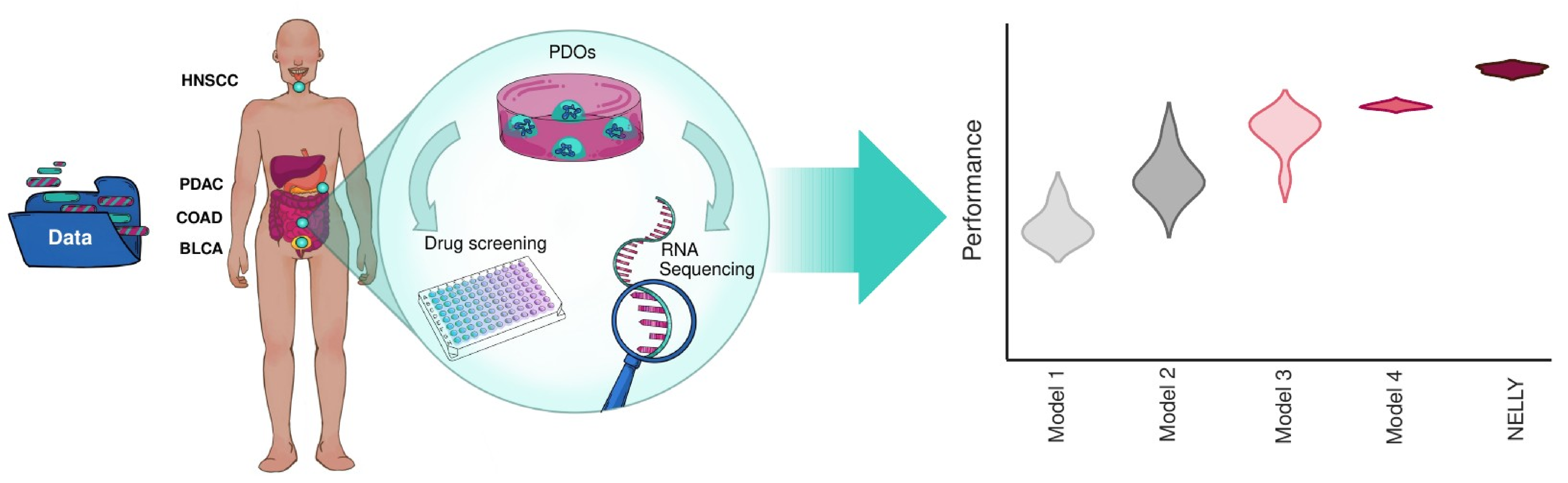

Figure 2. Assembly of a pan-cancer patient-derived organoid (PDO) dataset.
We assembled a PDO-based pharmacogenomic dataset and used it as an independent benchmark for zero-shot IC50 prediction. The PDO benchmarking dataset, includes 19 PDO lines from colorectal adenocarcinoma (COAD) [1], 11 PDO lines from bladder cancer (BLCA) [2], 5 PDO lines from pancreatic ductal adenocarcinoma (PDAC) [3] and 19 PDO lines from head and neck squamous cell carcinoma (HNSCC) [29, this study].

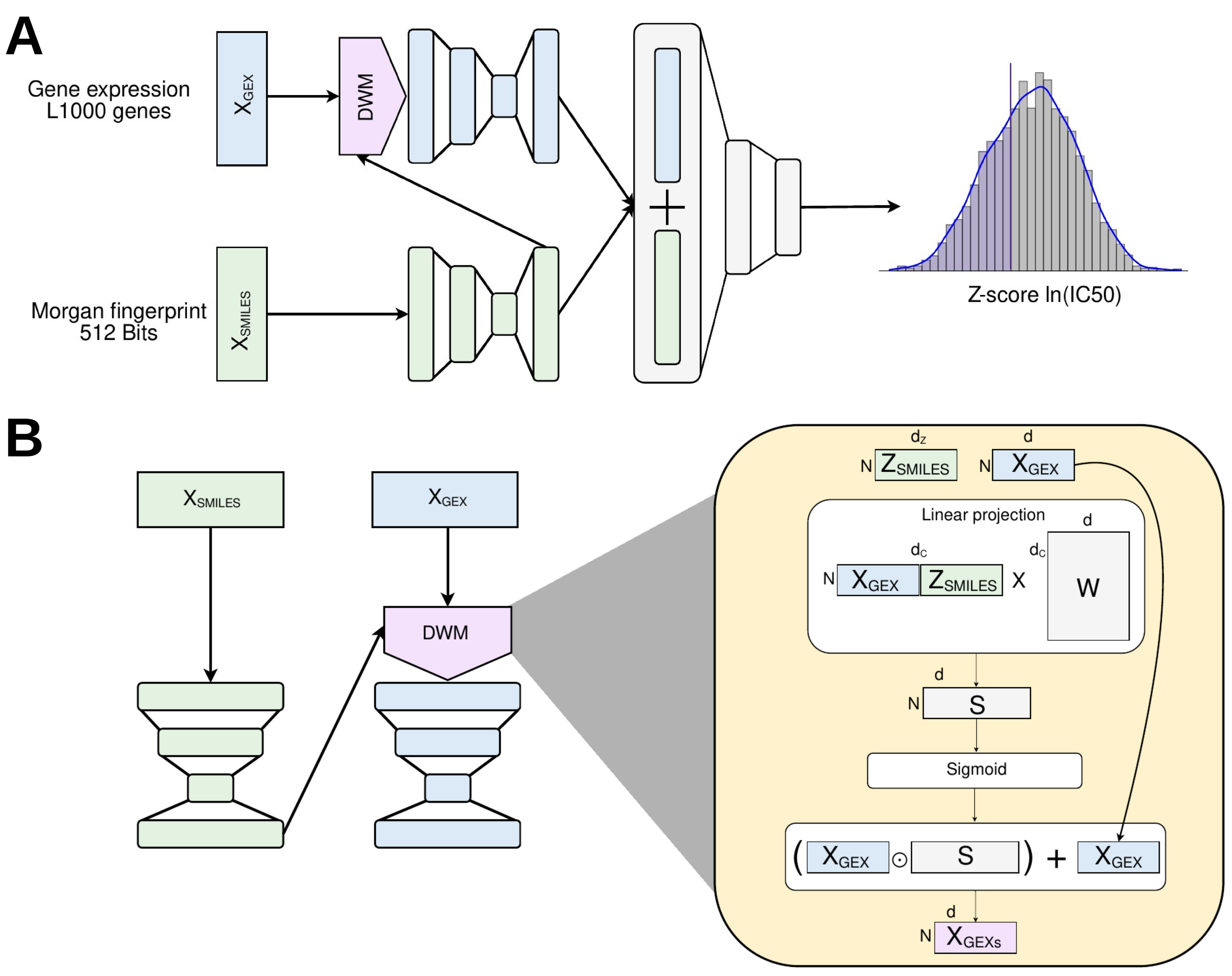

Figure 3. NELLY's architecture is designed to enhance model interpretability.
A. NELLY consists of a multimodal neural network with three dedicated heads. The first two heads extract informative patterns from gene expression (GEX) and SMILES into an embedding space. A third head, the prediction head, incorporates the GEX and SMILES embedding (Z_{SMILES}) to predict a Z-score ln(IC50) value.
B. Dynamic Weighting Mechanism (DWM). The DWM is a novel module design to enable model interpretability. It consists on a linear projection, followed by sigmoid scaling. The learned weights are used in an element-wise multipication on X_{GEX} acting as a gating mechanism.

3. Results

DWM-NELLY outperforms existing methods under fixed-line evaluation

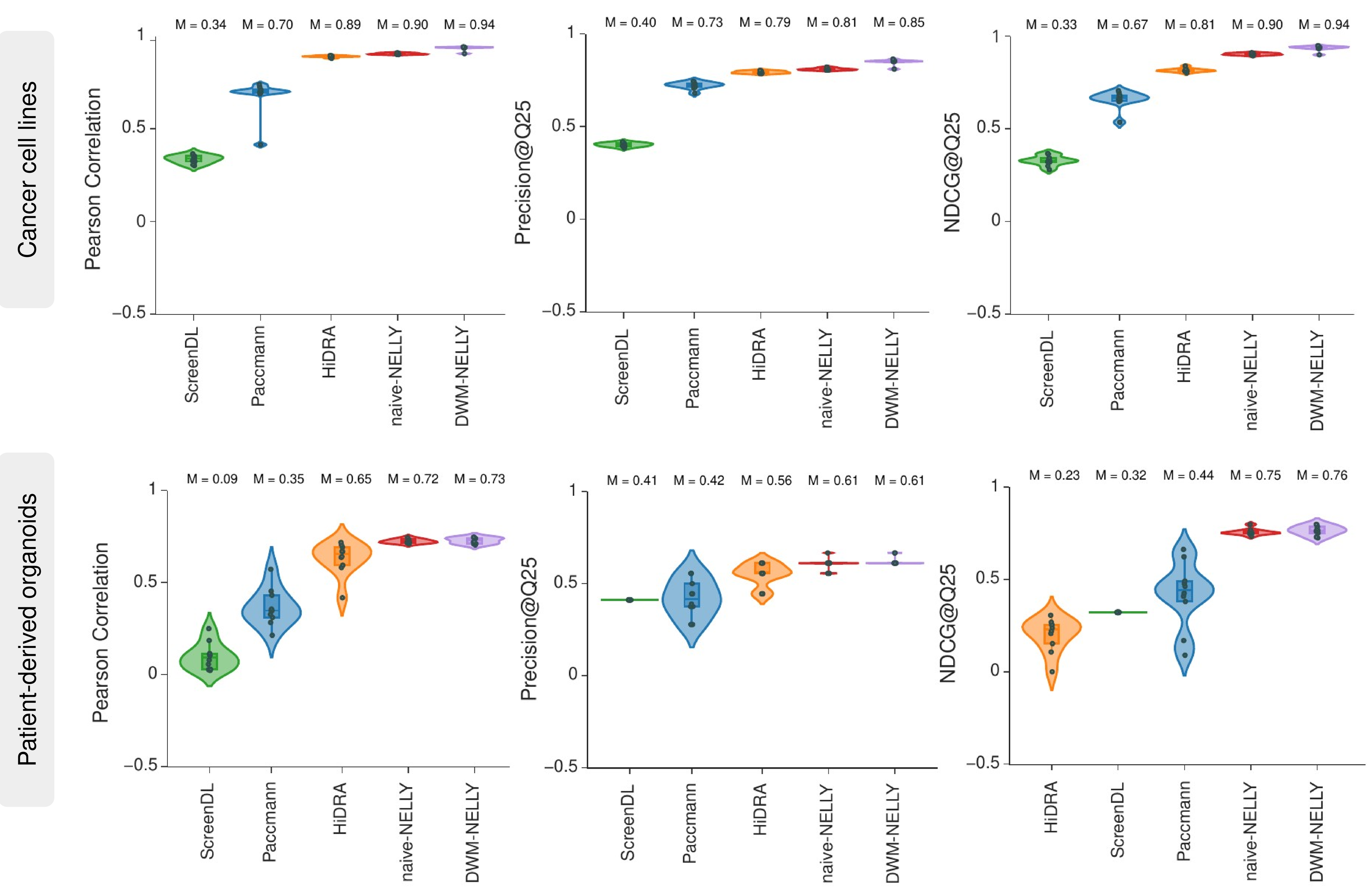

Figure 4. Robust benchmark for patient-centric drug prioritization.
Model performance is compared across a series of existing methods. Performance was assess under fixed-line evaluation using regression and ranking metrics.

DWM-derived weights can be leverage to study mechanism of action

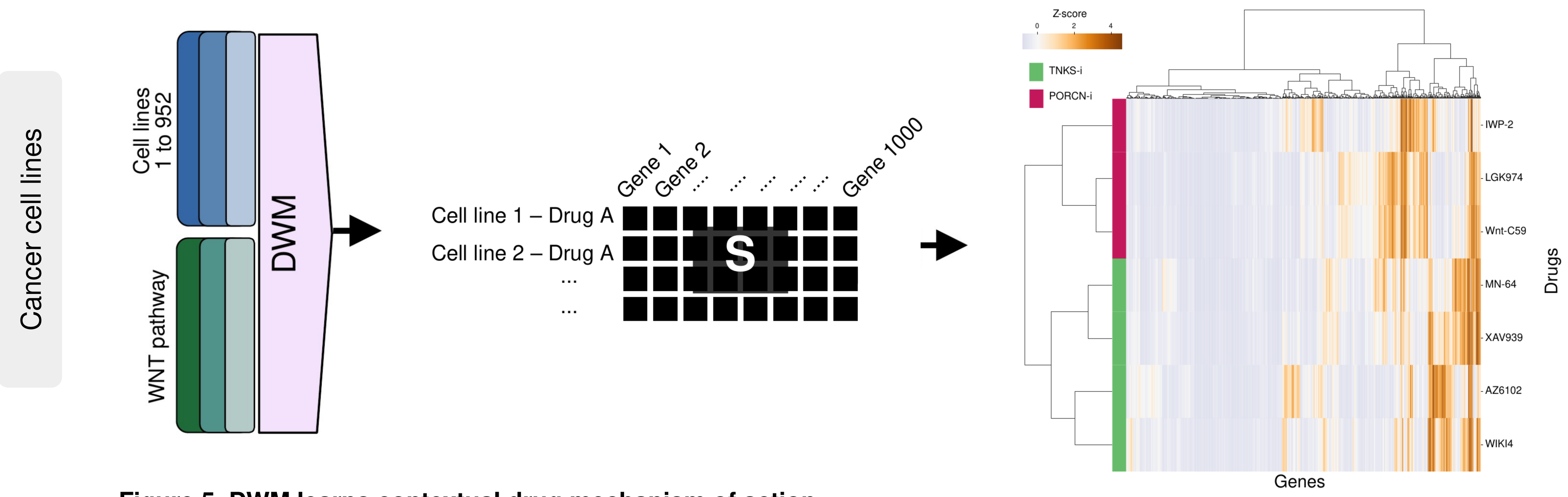

Figure 5. DWM learns contextual drug mechanism of action.
DWM-derived gene attribution matrix (*S*) was extracted for drugs targeting the WNT pathway. Hierarchical clustering of drug-averaged attribution matrix revealed mechanism of action groupings based on model-assigned gene weights for the two WNT inhibitors.

DWM-NELLY generated patient-specific drug recommendations in HNSCC PDOs

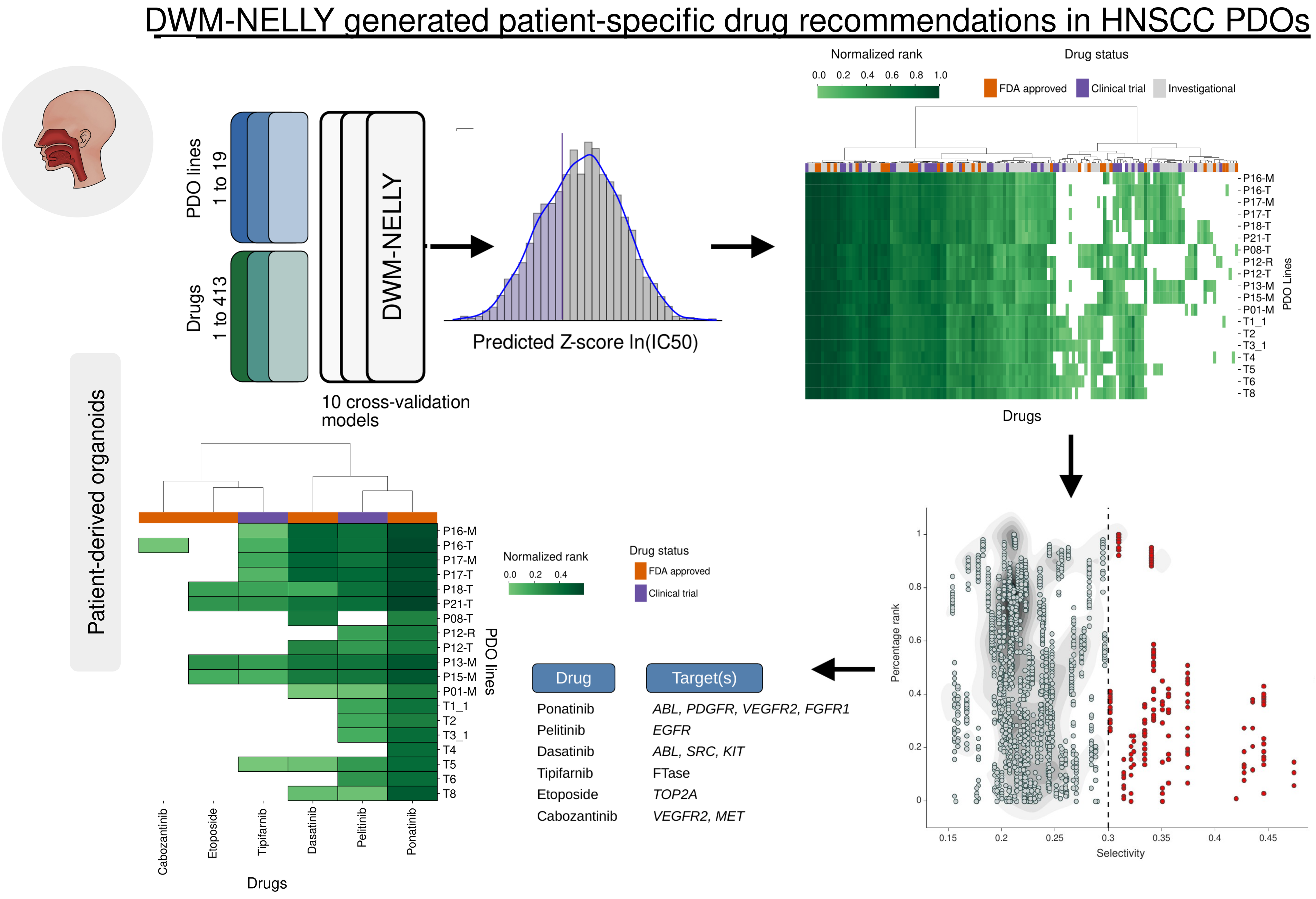

Figure 6. DWM-NELLY predict patient-specific drug recommendations in HNSCC.
Pre-trained DWM-NELLY was used to predict IC50 for our HNSCC PDO-cohort. For each organoid line, we then extracted the drug responses falling into the lower 25% of the predicted IC50 distribution. Predicted responses where transformed into a normalized rank score, from 0 to 1. Drug recommendations were further filtered to distinguish broad cytotoxicity from selective activity, generating patient-specific drug recommendations.

4. Conclusions

- Overall, existing methods achieve robust generalization to patient-derived cancer organoids.
- DWM-NELLY outperforms existing methods even when these are attention-based models.
- Dynamic Weighting Mechanism provides a simple but useful module to interpret model-derived weights.
- We provide the first pan-cancer PDO-based pharmacogenomics atlas.

5. Future work

- Future work should explore domain adaptation strategies, this may help improve the transfer of representations learned from cell lines to PDOs and other patient-derived systems.
- Investigate whether DWM-derived gene attribution can help to identify biomarkers of sensitivity or resistance for specific therapeutic agents.
- Investigate whether DWM-derived gene attribution can be leveraged to identify candidate combination therapies, in a patient-specific manner, by integration of graph-based approaches.

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